

The minimal polynomial

Example:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} ; \text{ch}_A(x) = (-1)^3(x-1)^3$$

$$B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} ; \text{ch}_B(x) = (-1)^3(x-1)^3$$

$$C = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} ; \text{ch}_C(x) = (-1)^3(x-1)^3$$

Note: (i) $A - 1 \cdot I_3 = 0$

(ii) $B - 1 \cdot I_3 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$\begin{bmatrix} 0 & 0 & 0 \end{bmatrix}$$

$$(B - I \cdot I_3)^2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

As a linear transformation,

$$e_1 \xrightarrow{B - I \cdot I_3} 0 \xrightarrow{B - I \cdot I_3} 0$$

$$e_2 \mapsto e_1 \mapsto 0$$

$$e_3 \mapsto 0 \mapsto 0$$

$$(iii) \quad (C - I \cdot I_3) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$e_1 \xrightarrow{C - I \cdot I_3} 0 \xrightarrow{C - I \cdot I_3} 0 \xrightarrow{C - I \cdot I_3} 0$$

$$e_2 \mapsto e_1 \mapsto 0 \mapsto 0$$

$$e_3 \mapsto e_2 \mapsto e_1 \mapsto 0$$

$$(C - I \cdot I_3)^3 = 0$$

Summary:

	A	B	C
<u>Char. poly.</u>	$(x-1)^3$	$(x-1)^3$	$(x-1)^3$
	$(A - 1 \cdot I_3)^3$	$(B - 1 \cdot I_3)^3$	$(C - 1 \cdot I_3)^3$
	$= 0$	$= 0$	$= 0$
	i.e.	i.e.	i.e.
	$\text{ch}_A(A)$	$\text{ch}_B(B)$	$\text{ch}_C(C)$
	$= 0$	$= 0$	$= 0$
$f(x) = x-1$	$A - 1 \cdot I_3$	$(B - 1 \cdot I_3)$	$(C - 1 \cdot I_3)$
	$= 0$	$\neq 0$	$\neq 0$
deg 1	i.e.	$(B - 1 \cdot I_3)^2$	$(C - 1 \cdot I_3)^2$
		$= 0$	$\neq 0$
	$f(A) = 0$	$f(B) \neq 0$	$(C - 1 \cdot I_3)^3 = 0$
		$f^2(B) = 0$	$f(C) \neq 0, f^4(C) \neq 0$
			$f^3(C) = \text{ch}_C(C)$
			$= 0$

Rank A, B, C have same characteristic polynomial

But different minimal polynomials

$$\min_A(x) = x-1$$

$$\min_B(x) = (x-1)^2$$

$$\min_C(x) = (x-1)^3 = \chi_C(x)$$

Defⁿ (1) $f(x) = a_m x^m + a_{m-1} x^{m-1} + \dots + a_0$
 $\in \text{Poly}(F)$

$$A \in M_{n \times n}(F)$$

$$f(A) = a_m A^m + a_{m-1} A^{m-1} + \dots + a_0 I_n$$

(2) The minimal polynomial

$\min_A(x) \in \text{Poly}(F)$ is

the non-zero polynomial of smallest degree s.t. $\min_A(A) = 0$

Rmk: $M_{n \times n}(F)$ is a finite dim'l
F-vector space of dimension n^2

$$\text{So } \{I_n, A, A^2, \dots, A^{n^2}\} \subseteq M_{n \times n}(F)$$

$n^2 + 1$

are linearly dependent

$$\Rightarrow \exists a_{n^2}, a_{n^2-1}, \dots, a_1, a_0 \in F$$

not all zero

s.t.

$$a_{n^2} A^{n^2} + a_{n^2-1} A^{n^2-1} + \dots + a_1 A + a_0 I_n = 0$$

$$\Rightarrow \text{If } f(x) = a_{n^2} x^{n^2} + \dots + a_1 x + a_0$$

$$\text{then } f(A) = 0$$

$$\Rightarrow \left\{ f(x) \in \text{Poly}(F) : \begin{array}{l} f(x) \text{ non-zero} \\ f(A) = 0 \end{array} \right\}$$

is non-empty

$\Rightarrow$ $\text{min}_A(x)$ exists

Well-ordering principle

Remark (2)

$$\left. \begin{array}{l} A \in M_{n \times n}(F) \\ f(x), \text{min}_A(x) \in \text{Poly}(F) \\ f(A) = 0 \end{array} \right\} \Rightarrow \begin{array}{l} f(x) \\ = \text{min}_A(x) q(x) \\ \text{for some } q(x) \\ \in \text{Poly}(F) \end{array}$$

Pf: Long division $\Rightarrow f(x) = \text{min}_A(x) q(x) + r(x)$

$\deg r(x) < \deg \text{min}_A(x)$

$$f(A) = \text{min}_A(A) q(A) + r(A)$$

$$0 = 0 + r(A)$$

$$\Rightarrow r(A) = 0$$

$$| \quad r(x) \neq 0$$

then $\bullet \deg r(x) < \deg \min_A(x)$
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$$\bullet r(A) = 0$$

contradict the defⁿ
 of $\min_A(x)$

$$\Rightarrow r(x) \geq 0 \Rightarrow f(x) = \min_A(x) q(x)$$

Example: $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $f(x) = (x-1)$

$$f^2(B) = (B - 1 \cdot I_3)^2 = 0$$

$$\Rightarrow f(x)^2 = \min_B(x) q(x)$$

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$$(x-1)^2 = \min_B(x) q(x)$$

$$\Rightarrow \min_B(x) \in \{(x-1), (x-1)^2\}$$

$$\text{But } B - I, \neq 0$$

$$\Rightarrow \min_B(x) \neq x-1.$$

$$\Rightarrow \min_B(x) = (x-1)^2.$$

Proposition : $A \in M_{n \times n}(\mathbb{C})$

TFAE: (1) A is diagonalizable

(2) $\mathbb{C}^n$ admits an eigenbasis for A

(3) $\min_A(x)$ has no repeated linear factors

$$\text{i.e. } \min_A(x) = \prod_{i=1}^r (x - \lambda_i)$$

for $\lambda_1, \dots, \lambda_r \in \mathbb{C}$ distinct

